

STAT 248 - Final Project Proposal

Akanshaa Borgohain

April 2026

1 The Question

Does an LSTM model trained on market microstructure features perform better than GARCH in predicting the next day's volatility regime (calm vs turbulent)?

Volatility measures how much an asset's price moves around over time. Markets do not experience constant and uniform volatility; rather they cluster into distinct regimes: calm periods ($VIX^1 \leq 20$) and turbulent periods ($VIX \geq 30$). The use of GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model to predict stock price volatility has been the industry standard since the 1980s. This project asks whether an LSTM (Long-Short-Term Memory) model, which can capture nonlinear and long memory dependencies, meaningfully outperforms GARCH. This analysis will span years 2005-2024, capturing 2008 Global Financial Crisis, 2020 COVID-19 crash and 2022 bear market.

2 Description of the data

This project will use six daily financial and macroeconomic time series sourced from the **Federal Reserve Bank of St. Louis Economic Data (FRED)** database via `fredapi` Python library, with S&P 500 prices from **Stooq** due to FRED's 10-year licensing restriction on that series. The six series are shown in Table 1.

Series	Source/FRED Code	Description
VIX	VIXCLS	CBOE Implied Volatility Index
S&P 500 Price	Stooq website	Index closing value
10-Year Treasury Yield	DGS10	Risk-free rate
2-Year Treasury Yield	DGS2	Short-end rates
Fed Funds Rate	DFF	Monetary policy signal
Credit Spread (BAA-AAA)	BAA10Y	Corporate stress indicator

Table 1: Data Series

All series have been acquired, and they are at **daily temporal resolution**, covering **January 2005 - December 2024**, which is approximately 5000 trading observations per series, for a total of 6 distinct time series. I will also derive two additional features: daily log return of the S&P 500 and a 21-day rolling realized volatility, bringing it to a set of 8 series.

3 Why is this a time series problem

Volatility exhibits strong temporal dependence, where large price moves tend to follow large moves, and calm periods tend to persist. This means that the observations are not independent, today's VIX level is highly predictive of tomorrow's. Treating this as an i.i.d observation would produce wrong models, invalid standard errors, and leakage if train/test splits are not done chronologically. Furthermore, whether the market is

¹CBOE Volatility Index- also known as fear index

entering a crisis or exiting one depends on the sequence of days preceding it. Therefore, time series methods are necessary to answer this question.

4 Proposed methods

From the course material, I will use three main methods. First, I will apply a GARCH(1,1) model (related to AR models and nonlinear regression). This will serve as the baseline model, fitting conditional variance as a function of past squared returns and past variance. Second, an LSTM(recurrent neural network) will be trained on a sliding window of the 8 features to predict next day's regime. Third, time series cross-validation(walk-forward expanding window) will be used to evaluate both models- standard k-fold cross-validation is invalid here because it breaks the temporal order and will cause data leakage.

5 Expected outcomes

A positive result would show that LSTM performs with higher precision, especially around the crisis onset, where early warning is important. A null result would show that there's not much difference in how GARCH and LSTM performs. Either outcome is scientifically informative: a null result would constitute modest evidence in favor of the efficient market hypothesis, while a positive result would suggest exploitable nonlinear regime structure in daily volatility dynamics.

References

- [1] Riechie et al. (2026). *A comparative study of LSTM and GARCH in predicting stock prices volatility and estimating value-at-risk*. In *AIP Conference Proceedings*, Vol. 3326, No. 1. AIP Publishing LLC.
- [2] Gratton, P. (2025). *Cboe Volatility Index (VIX) or the Fear Index: Explanation and Calculation*. Investopedia. Retrieved from <https://www.investopedia.com/ask/answers/021015/what-cboe-volatility-index-vix.asp>
- [3] The Investopedia Team. (2024). *Understanding GARCH Models: Volatility Prediction in Finance*. Investopedia. Retrieved from <https://www.investopedia.com/terms/g/garch.asp>